BALAJI DHANAVANTHAN

SENIOR DATA ENGINEER | DATA & ANALYTICS ENGINEERING

Chennai, India | Open to Global Opportunities

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PROFESSIONAL SUMMARY

Azure Data Engineer with 4.5+ years of experience building cloud data pipelines and Lakehouse solutions across enterprise technology, manufacturing, and sports/entertainment domains. Hands-on with Azure Data Factory, Azure Databricks, PySpark, SQL, Delta Lake, ADLS Gen2, dbt, and AWS analytics services. Experienced in metadata-driven ingestion, Medallion Architecture, orchestration, data validation, CI/CD, performance tuning, and AI-assisted workflow automation using Databricks Genie.

TECHNICAL SKILLS

Cloud: Azure, AWS

ETL / ELT: Azure Data Factory (ADF), Databricks, dbt, Snowflake

Programming: Python, PySpark, SQL, Spark SQL

Data Platforms: ADLS Gen2, Delta Lake, Azure SQL Database, Amazon Redshift

Architecture & Modeling: Lakehouse, Medallion Architecture, Data Warehousing

Orchestration & Monitoring: Databricks Jobs, Databricks Workflows, Apache Airflow, Azure Monitor

Automation: Databricks Genie, Power Automate

DevOps & CI/CD: Azure DevOps, GitHub, CI/CD

PROFESSIONAL EXPERIENCE

EY (Ernst & Young) | Chennai, India — Oct 2025 - Present

Senior Analyst | Azure Data Engineer

Enterprise Technology Platform | Lakehouse Engineering & AI-Assisted Data Automation

- **Lakehouse Platform:** Engineered a cloud Lakehouse using Azure Data Factory, Databricks Auto Loader, Delta Lake, ADLS Gen2, and Power Automate to standardize ingestion and transformation across enterprise data sources.
- **Metadata-Driven Ingestion:** Established a reusable, configuration-driven ADF framework for files, APIs, and databases; project results reported a 70% reduction in new-source onboarding effort.
- **Medallion Architecture:** Structured Bronze, Silver, and Gold layers processing 100K+ records per day, improving consistency and preparing curated datasets for downstream analytics.
- **PySpark Transformations:** Built reusable transformation frameworks and business-rule logic for recurring reports and complex requirements; project results reported 60% faster reporting turnaround.
- **Genie Data Agents:** Created Databricks Genie data agents grounded in approved Delta tables, table/column descriptions, and business terminology, enabling natural-language exploration of curated operational data.
- **AI-Assisted Automation:** Used Genie to generate or refine scripts and streamline repetitive manual tasks, reviewing outputs before use; project results reported a 30% improvement in business-user productivity.
- **Orchestration & Monitoring:** Coordinated ADF pipelines with Databricks Jobs and Workflows and reviewed Azure Monitor signals and execution logs to investigate failures and improve operational visibility.
- **Data Validation & Delivery:** Applied validation and reconciliation checks across curated datasets and collaborated with engineers, analysts, architects, and business stakeholders across teams of up to 24 members.

Systech Solutions Private Limited | Chennai, India — May 2022 - Sep 2025

Data Engineer | Role progression from Data Engineer Trainee

Manufacturing | TDS Lithium Ion Battery Gujarat Pvt. Ltd. | Production Data Integration — Aug 2023 - Sep 2025

- **ETL / ELT Pipeline Engineering:** Delivered production pipelines using ADF and Databricks for SAP-to-Azure SQL loads and CSV, Excel, text, and Parquet sources, applying repeatable ingestion and SCD-based transformations.
- **Data Accuracy & Validation:** Maintained reported 99.99% data accuracy with Python-based tests and validation checks across manufacturing data before downstream reporting.
- **SQL Performance Tuning:** Improved Azure SQL query performance by up to 40% through query tuning, indexing strategies, and data-model optimization for frequently used production and reporting queries.

PROFESSIONAL EXPERIENCE (CONTINUED)

Systech Solutions Private Limited | Data Engineer (continued)

Manufacturing | TDS Lithium Ion Battery Gujarat Pvt. Ltd. (continued)

- **CI/CD & Monitoring:** Configured CI/CD delivery using Azure DevOps and GitHub; used Power Automate and SQL-based logging to improve run visibility and reduce monitoring/support effort by a reported 85%.
- **Data Modeling & Reporting:** Prepared source-to-target mapping (STTM), dimensional models, and business data marts; automated recurring stakeholder reporting workflows using ADF and Power Automate.

Sports & Entertainment | Fox Corporation | Media Analytics Data Platform Oct 2022 - Jul 2023

- **Cloud Data Pipelines:** Built cloud ingestion and transformation pipelines with Databricks, dbt, Apache Airflow, AWS, and Amazon Redshift for recurring sports and entertainment reporting.
- **Multi-Source Integration:** Integrated Nielsen, Amagi, FAST channels, APIs, and third-party platforms, standardizing source data into consistent datasets for analytics and reporting workflows.
- **dbt & JSON Transformation:** Applied modular dbt business logic and reusable JSON-flattening routines to convert nested payloads into structured, analytics-ready data.
- **Deployment Automation:** Streamlined deployment of Databricks, Airflow, and dbt assets using GitHub Actions and CI/CD practices, improving release consistency and version control.
- **Redshift Development:** Authored Amazon Redshift stored procedures and optimized recurring transformation workloads to support consolidated reporting.

Initial Assignment | Data Engineer Trainee May 2022 - Sep 2022

- **Pipeline Foundations:** Worked with Databricks, SQL, Python, dbt, and Snowflake to establish ETL/ELT workflows and build a foundation in cloud data transformation.
- **Source Integration:** Configured source ingestion and applied business transformation rules to prepare structured datasets for reporting and downstream consumption.
- **Testing & Validation:** Wrote SQL and Python test scripts to validate transformed data, catch inconsistencies, and improve confidence in pipeline outputs.
- **BI & Project Support:** Produced Power BI dashboards for KPI visibility and stakeholder ticket tracking, supporting project reporting needs.
- **Issue Resolution:** Resolved JIRA issues with the project team to support dependable data operations and timely delivery.

LEADERSHIP & KNOWLEDGE SHARING

Learning & Development Champion | Systech Solutions | 2023 - 2025

- **Mentoring:** Guided interns and new joiners on Azure SQL, Azure Data Factory, and data warehousing to support onboarding.
- **Technical Workshops:** Conducted three hands-on Azure and SQL workshops, using practical demonstrations to share data-engineering knowledge.

CERTIFICATIONS

- Databricks Certified Data Engineer Associate
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- Databricks Lakehouse Fundamentals
- dbt Fundamentals
- GitHub Copilot Fundamentals

EDUCATION

Bachelor of Engineering, Electrical & Electronics Engineering 2015 - 2019
Saveetha Engineering College